

The nature of crystalline states

There're four types of crystalline solids.

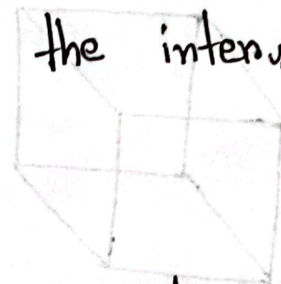
1. Ionic
2. Covalent
3. Metallic
4. Molecular

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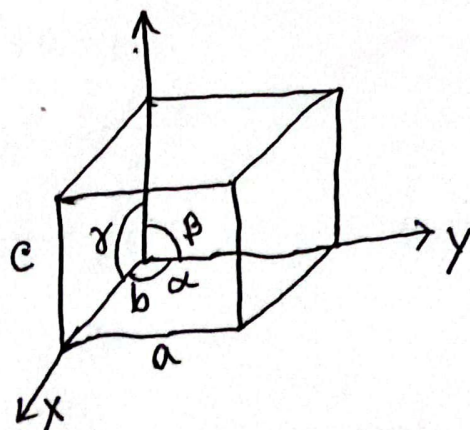
Crystals are bounded by a number of surface (usually flat) which are called faces.

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An edge is formed by the intersection of any two adjacent faces.



An interfacial angle is formed by the intersection of three or more faces. Ex- α, β, γ



$a, b, c \rightarrow$ length/edge
 $\alpha, \beta, \gamma \rightarrow$ interfacial angle

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Space lattice: The regular and periodic arrangement of atoms, ions and molecules at the lattice point of a crystal in three dimensional phase is called space lattice/crystal lattice.

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All space lattice are crystal lattice but all crystal lattice are not space lattice.

Unit cell: A unit cell is the smallest building blocks of crystal structure, repetition of the unit cell forms a single solid crystal.

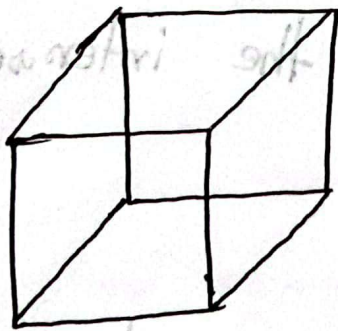
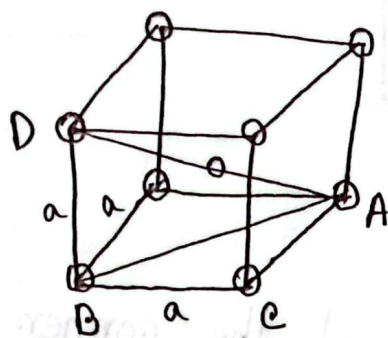


Fig 01: A unit cell



Construction of Crystal

BCC



Atoms are located at the corner and at the center.

At the corner,

$$\text{atoms} = 8 \times \frac{1}{8} = 1$$

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At the center,

$$\text{atoms} = 1$$

$$\therefore \text{equivalent \# number of atoms} = 1 + 1 = 2$$

$$\therefore \text{Co ordination number} = 8$$

Example: Na, K, W, Fe, Li, Rb.

Here, in $\triangle ABC$ triangle,

$$BC^2 + AC^2 = AB^2$$

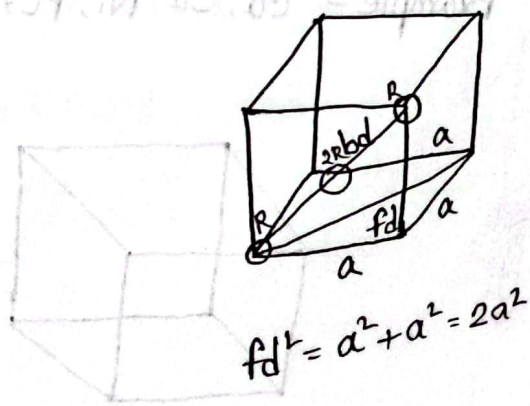
$$f_d^2 + a^2 = bd^2$$

$$\Rightarrow 2a^2 + a^2 = (4R)^2$$

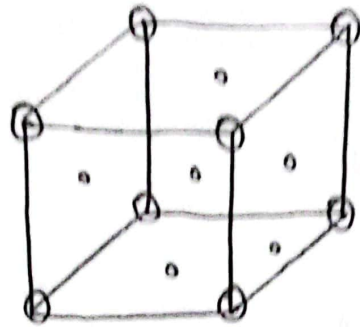
$$\Rightarrow 3a^2 = (4R)^2$$

$$\Rightarrow \sqrt{3}a = 4R$$

$$\Rightarrow a = \frac{4R}{\sqrt{3}}$$



fcc



Atoms are located at the corners and at the center of each face.

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At the corner,

$$\text{atom} = 8 \times \frac{1}{8} = 1$$

At the center,

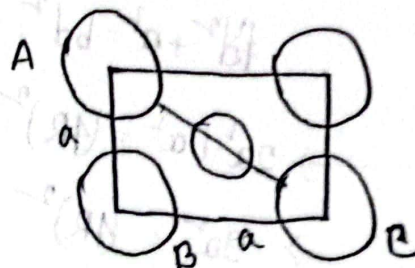
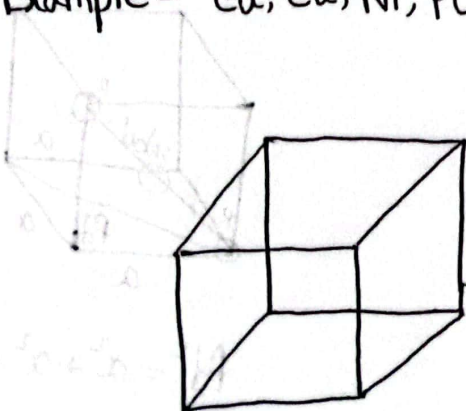
$$\text{atom} = 6 \times \frac{1}{2} = 3$$

number of atoms = 1 + 3 = 4

coordination number = 12

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Example - Ca, Cu, Ni, Pt, Ag, Au, Al



Hence, in $\triangle ABC$ triangle,

$$AB^2 + BC^2 = AC^2$$

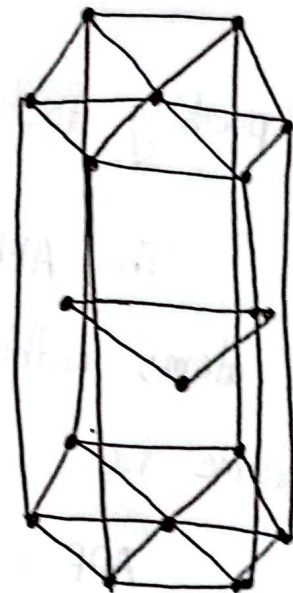
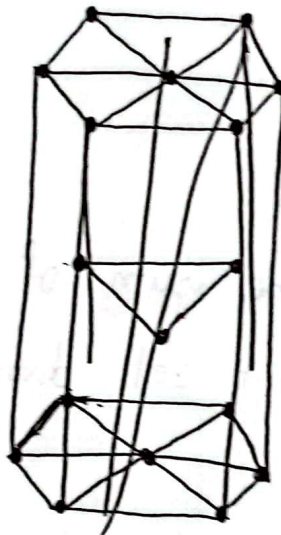
$$\Rightarrow a^2 + a^2 = (4R)^2$$

$$\Rightarrow 2a^2 = (4R)^2$$

$$\Rightarrow a = \frac{4R}{\sqrt{2}}$$

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Hcp



Atoms are located at the corner and at the center.

At the corner,

$$\text{atoms} = 2 \times \left(6 \times \frac{1}{6}\right) = 2$$

At the center,

$$\text{atoms} = 3$$

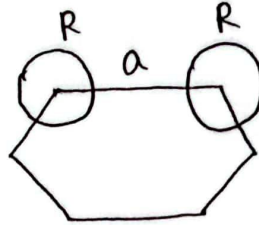
At the faces,

$$\text{atoms} = 2 \times \frac{1}{2} = 1$$

equivalent number of atoms = $2 + 3 + 1 = 6$

Coordination number = 12

Ex- Mg, Zn, Cu, Ti, Cd, Co



$$a = 2R$$

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Atomic packing factor (APF)

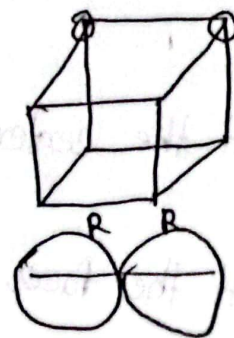
The APF is the sum of the sphere volume of all atoms within a unit cell divided by the unit cell volume i.e.,

$$APF = \frac{\text{volume of atoms in a unit cell}}{\text{total unit cell volume}}$$

Atomic packing factor calculation for -

1. Simple Cubic structure (sc)

$$\begin{aligned} APF_{sc} &= \frac{1 \times \frac{4}{3}\pi R^3}{a^3} \\ &= \frac{4\pi R^3}{3a^3} = \frac{4\pi R^3}{3(2R)^3} \\ &= 0.52 = 52\% \end{aligned}$$



each atom = 1

2. Body centered cubic structure (BCC)

$$\begin{aligned} \text{APF}_{\text{BCC}} &= \frac{2 \times \frac{4}{3} \pi R^3}{a^3} \\ &= \frac{8 \pi R^3}{3 \times \left(\frac{4R}{\sqrt{3}}\right)^3} \\ &= \frac{8 \pi R^3 \times 3\sqrt{3}}{3 \times 64 R^3} \\ &= \frac{\sqrt{3} \pi}{8} \\ &= 0.68 \\ &= 68\% \end{aligned}$$

eq no of atoms = 2

$$a = \frac{4R}{\sqrt{3}}$$

Fig —

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3. Face centered cubic structure (FCC)

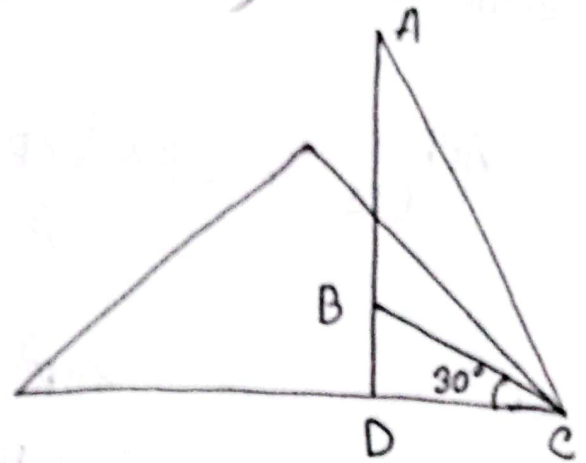
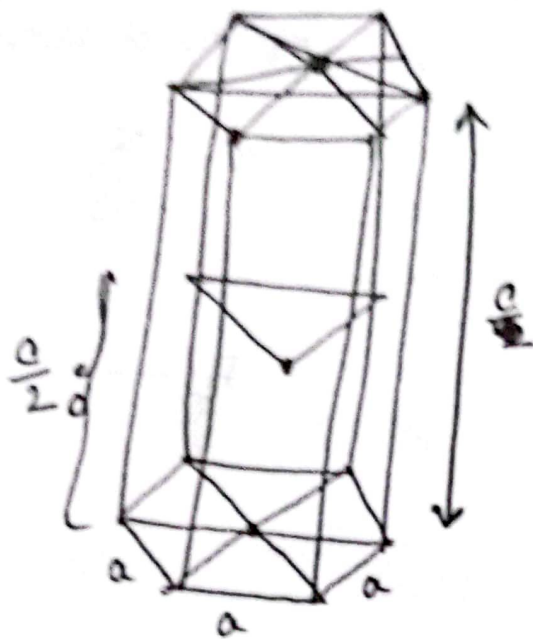
$$\begin{aligned} \text{APF}_{\text{FCC}} &= \frac{4 \times \frac{4}{3} \pi R^3}{a^3} \\ &= \frac{\frac{16}{3} \times \pi R^3}{\left(\frac{4R}{\sqrt{2}}\right)^3} \\ &= \frac{16 \pi R^3 \times 2\sqrt{2}}{3 \times 64 R^3} \\ &= \frac{\pi}{3\sqrt{2}} = 0.74 = 74\% \end{aligned}$$

eq no of atoms = 4

$$a = \frac{4R}{\sqrt{2}}$$

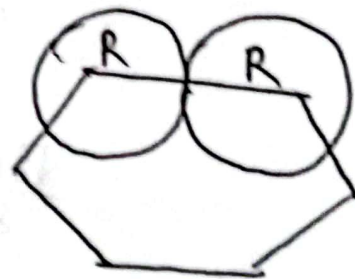
Fig —

4. Hexagonal close packed structure (HCP)



$$a = 2R$$

$$\Rightarrow R = \frac{a}{2}$$



In $\triangle BCD$ triangle,

$$\cos 30^\circ = \frac{CD}{BC}$$

$$\Rightarrow BC = \frac{R}{\frac{\sqrt{3}}{2}}$$

$$\Rightarrow BC = \frac{2R}{\sqrt{3}}$$

$$\Rightarrow BC = \frac{2 \times \frac{a}{2}}{\sqrt{3}}$$

$$\Rightarrow BC = \frac{a}{\sqrt{3}}$$

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In $\triangle ABC$ triangle,

$$AC^2 = AB^2 + BC^2$$

$$\Rightarrow a^2 = \frac{c^2}{4} + \frac{a^2}{3}$$

$$\Rightarrow \frac{c^2}{4} = a^2 - \frac{a^2}{3} = \frac{2a^2}{3}$$

$$\Rightarrow \frac{a^2}{c^2} = \frac{3}{8}$$

$$\Rightarrow \frac{a}{c} = \sqrt{\frac{3}{8}}$$

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$\therefore$ Total volume of HCP = Area of the base $\times$ height

$$= 6 \times \frac{\sqrt{3}}{4} \times a^2 \times c$$

$$= \frac{3\sqrt{3}}{2} a^2 c$$

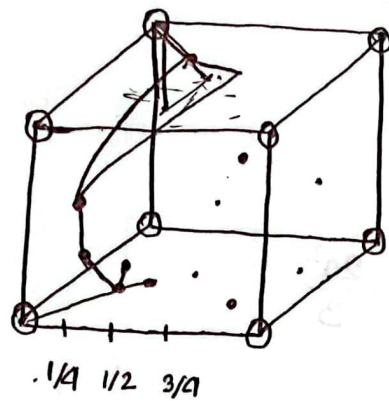
$$\therefore APF_{HCP} = \frac{6 \times \frac{4}{3} \pi R^3}{\frac{3\sqrt{3}}{2} a^2 c}$$

$$= \frac{8\pi \times \left(\frac{a}{2}\right)^3 \times 2}{3\sqrt{3} a^2 c}$$

$$= \frac{16\pi a^3}{24\sqrt{3} a^2 c}$$

$$= \frac{2\pi}{3\sqrt{3}} \times \frac{a}{c} = \frac{2\pi}{3\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{8}} = 0.74 \approx 74\%$$

Diamond structure



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equivalent number of atoms $(8 \times \frac{1}{8} + \frac{1}{2} \times 6 + 4)$

$$= 1 + 3 + 4$$

$$= 8$$

Two types of atom

i) Extension atoms $(8 \times \frac{1}{8} + \frac{1}{2} \times 6) = 4$

ii) Interior atoms = 4

APF

Diamond

$$= \frac{8 \times \frac{4}{3} \pi R^3}{a^3}$$

$$R = \frac{\sqrt{3}a}{8}$$

$$= \frac{8 \times \frac{4}{3} \times \left(\frac{\sqrt{3}a}{8}\right)^3}{a^3}$$

$$\Rightarrow a = \frac{8R}{\sqrt{3}}$$

$$= 0.34 = 34\%$$

Coordination number = 4

DATE: / /

Properties

1. Hard in nature
2. Low coefficient of friction
3. High thermal conductivity
4. High electrical resistivity
5. Low thermal expansion coefficient
6. High strength
7. Chemical corrosion resistance
8. Optically transparent (UV, IR)
9. Biologically compatible
10. High Melting point (3550°C)
11. High boiling point (4827°C)

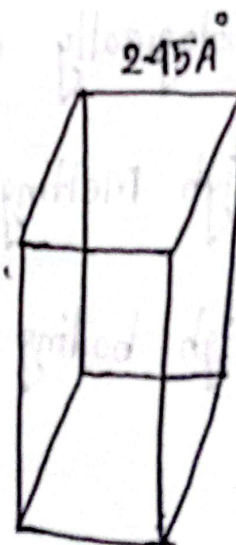
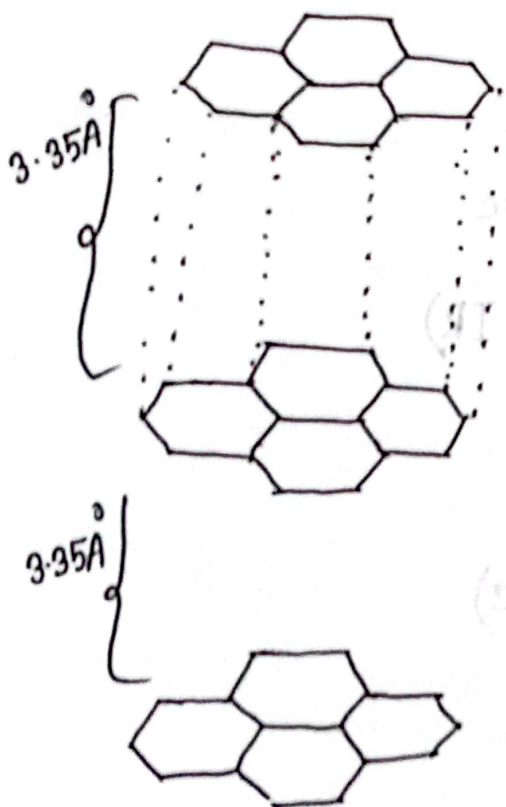
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Applications

1. Cutting tools
2. Jewellery
3. Semi conductor device
4. Optical recording disc
5. Microscopic tools

Graphite structure

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monoclinic crystal system

Properties

1. High thermal conductivity
2. High electric conductivity
3. Highly flexible
4. High hardness
5. 200 times stronger than steel
6. can exposed generate electricity by exposed sunlight.
7. Transparent in nature
8. Anti bacterial effect

Applications

1. pencil
2. Lubricant
3. Air craft engine, rocket
4. Bearing
5. Powder metallurgy

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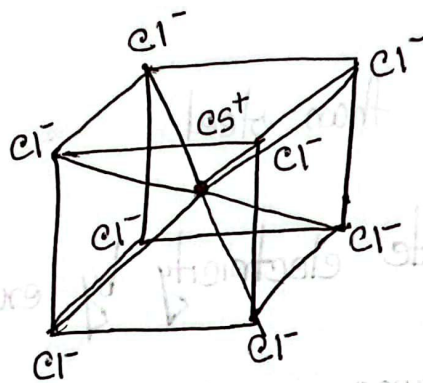
6. Grinding wheel

7. paddle blade

8. Solar cell (Graftin)

9. Storage cell

CsCl structure

Anion Cl^-
Cation Cs^+

Coordination number = 8

Properties

1. Soluble in water

2. Ionic compound

3. Cations are located in centre

4. Anions are located in corners

5. has BCC structure.

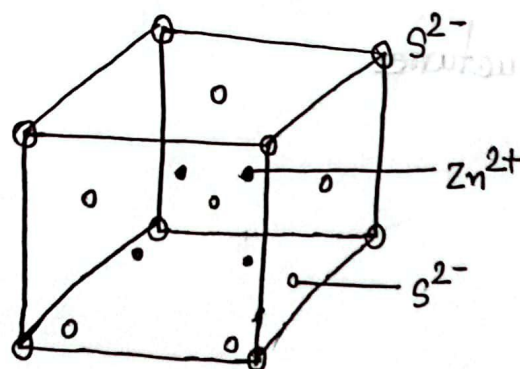
6. Melting point is 645°C

7. Boiling point is 1300°C

Applications

1. Centrifuge machine
2. Radiotherapy
3. Electrical conducting glass

ZnS/zinc blende (FCC)



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equivalent number of atoms

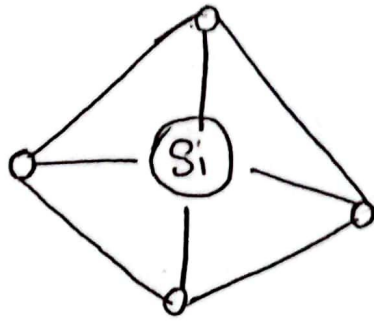
$$8 \times \frac{1}{8} + 6 \times \frac{1}{2} + 4 = 8$$

Same structure as ZnTe, SiC, CuBr, CuI, CuF

Applications

1. Painting
2. Plastic and Rubber industry

Silicate/ SiO_4

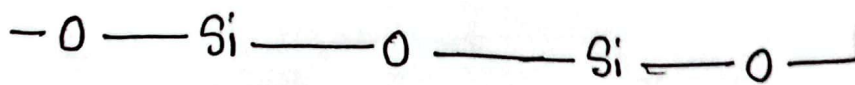


Tetra-hedron structure

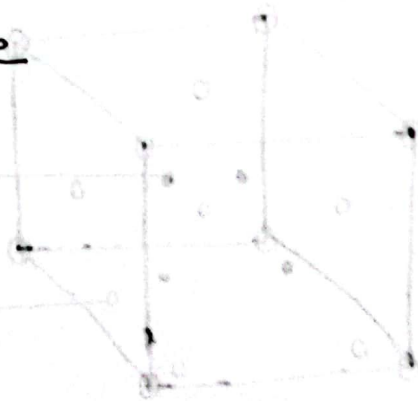
50% ionic

50% covalent

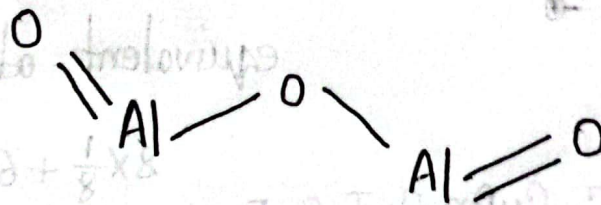
Silica/ SiO_2 /Quartz



Tetrahedral structure



Alumina/ Al_2O_3



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#

CN

r_c/r_a

shape

2

< 0.195



3

$0.155 - 0.225$



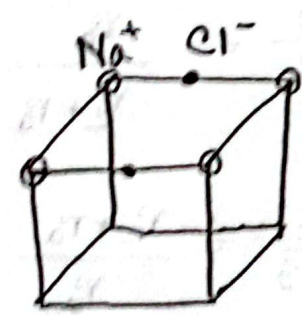
4

$0.225 - 0.414$



6

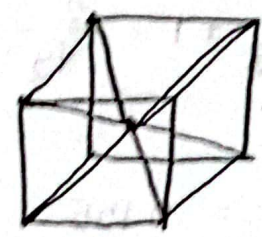
$0.414 - 0.732$



— NaCl (FCC)

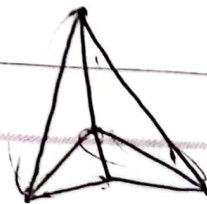
8

$0.732 - 1$



— CsCl (BCC)

Sub.:

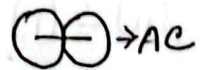
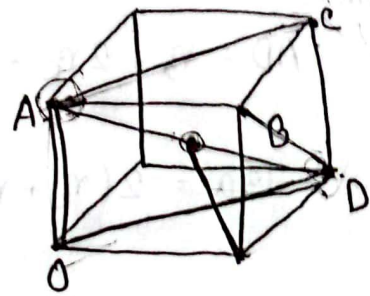


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CN - 4

$$AB = BC = CD = a$$

$$\begin{aligned} AC &= \sqrt{AD^2 + CD^2} \\ &= \sqrt{a^2 + a^2} = \sqrt{2} a \end{aligned}$$



Again,

In $\triangle AOD$ triangle,

$$AO^2 + OD^2 = AD^2$$

$$\Rightarrow a^2 + (\sqrt{2}a)^2 = AD^2$$

$$\Rightarrow AD^2 = 3a^2$$

$$\Rightarrow AD = \sqrt{3} a$$

Now,

$$AC = r_a + r_a$$

$$\Rightarrow \sqrt{2} a = 2r_a$$

$$\Rightarrow r_a = \frac{a}{\sqrt{2}} \text{ --- (i)}$$

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Further,

$$AD = r_c + 2r_c + r_a$$

$$\Rightarrow \sqrt{3}a = 2(r_c + r_a)$$

$$\Rightarrow r_c + r_a = \frac{\sqrt{3}a}{2} \quad \text{--- (ii)}$$

$$\therefore (ii) \div (i)$$

$$\frac{\frac{\sqrt{3}a}{2}}{\frac{a}{\sqrt{2}}} = \frac{r_c + r_a}{r_a}$$

$$\Rightarrow \frac{\sqrt{3}a}{2} \times \frac{\sqrt{2}}{a} = 1 + \frac{r_c}{r_a}$$

$$\Rightarrow \frac{r_c}{r_a} = \frac{\sqrt{3}}{\sqrt{2}} - 1$$

$$\Rightarrow \frac{r_c}{r_a} = 0.224$$

$$\therefore \frac{r_c}{r_a} = 0.224$$

$\rightarrow$

$$AB = BC = a$$

$$AC = \sqrt{2} a$$

In $\triangle ABC$ triangle,

$$AC = \sqrt{2} a$$

$$\Rightarrow 2(r_c + r_a) = \sqrt{2} a$$

$$\Rightarrow r_c + r_a = \frac{a}{\sqrt{2}} \quad \text{--- (i)}$$

Again,

$$AB = a = 2r_a$$

$$\Rightarrow r_a = \frac{a}{2} \quad \text{--- (ii)}$$

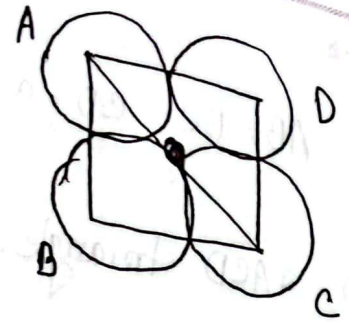
(ii) $\div$ (i)

$$\frac{r_c + r_a}{r_a} = \frac{\frac{a}{\sqrt{2}}}{\frac{a}{2}}$$

$$\Rightarrow \frac{r_c}{r_a} + 1 = \frac{a \times 2}{a \times \sqrt{2}}$$

$$\Rightarrow \frac{r_c}{r_a} = \sqrt{2} - 1 = 0.414$$

$$\therefore \frac{r_c}{r_a} = 0.414 R$$



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CN-8

$$AB = BC = CD = a$$

In $\triangle ACD$ triangle.

$$AD = \sqrt{3} a$$

$$\Rightarrow 2(r_a + r_c) = \sqrt{3} a$$

$$\Rightarrow r_c + r_a = \frac{\sqrt{3}}{2} a \quad \text{--- (i)}$$

Again,

$$AB = 2r_a$$

$$\Rightarrow a = 2r_a$$

$$\Rightarrow r_a = \frac{a}{2} \quad \text{--- (ii)}$$

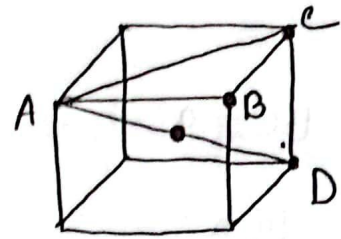
(i) $\div$ (ii)

$$\frac{r_c + r_a}{r_a} = \frac{\sqrt{3}}{2} a \times \frac{2}{a}$$

$$\Rightarrow \frac{r_c}{r_a} + 1 = \sqrt{3}$$

$$\Rightarrow \frac{r_c}{r_a} = \sqrt{3} - 1 = 0.732$$

$$\therefore \frac{r_c}{r_a} = 0.732$$



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